

Customer Purchase Propensity

Data Cleaning and Feature Engineering Pipeline
Project Planning & Final Summary

Step 1: Project Planning and Problem Framing

What is Data Analysis?

Data Analysis is the process of collecting, cleaning, transforming, and analyzing data to discover useful information and support decision-making.

Steps in a Data Science Project

- Problem Understanding
- Data Collection
- Data Cleaning and Preprocessing
- Exploratory Data Analysis
- Feature Engineering
- Model Building
- Model Evaluation
- Deployment and Monitoring

Machine Learning Problem

This dataset can be framed as a **Binary Classification** problem.

Target Variable: purchased

- purchased = 1 → Customer made a purchase
- purchased = 0 → Customer did not make a purchase

The future Machine Learning model can use customer demographics, transaction history, product information, and engineered features to predict purchase behavior.

Final Summary

Techniques Used

- Loaded data from CSV, JSON, SQL, and API sources.
- Merged datasets using customer_id and product_id.
- Performed Univariate, Bivariate, and Multivariate EDA.
- Handled missing data using Simple Imputer, Most Frequent Imputation, Missing Indicator, Random Sample Imputation, KNN Imputer, MICE, and Complete Case Analysis.
- Detected outliers using Z-score, IQR, and Percentile methods.
- Converted date variables and created date-based features.
- Applied Label Encoding, One Hot Encoding, and Ordinal Encoding.
- Applied StandardScaler, MinMaxScaler, MaxAbsScaler, RobustScaler, and Normalizer.
- Created interaction features and transformed numerical features.
- Performed binning and created the frequent_buyer feature.

Biggest Issues in Raw Data

- Missing values
- Outliers
- Different data sources
- Categorical variables
- Date and time variables
- Mixed data types
- Skewed numerical variables

Best Imputation and Scaling

Median and Most Frequent Imputation are simple and useful for general preprocessing. KNN and MICE can provide better estimates when variables are correlated.

RobustScaler is useful when outliers are present because it uses the median and interquartile range.

StandardScaler is useful for normally distributed numerical features.

Conclusion

The final dataset was cleaned, transformed, encoded, scaled, and feature-engineered. The processed dataset is ready for future Machine Learning Binary Classification.